

Vibe Shuffle: Browser-Based Affective Music Selection from Facial Valence and Cardiac Arousal

Markus Baumann

Ludwig-Maximilians-Universität München

Munich, Germany

markus.baumann@campus.lmu.de

Abstract

Music is widely used to regulate emotion, yet mainstream recommenders optimise for long-run engagement rather than a listener’s momentary affective state. We present *Vibe Shuffle*, a client-only web application that estimates a listener’s valence and arousal in real time—valence from facial expression and arousal from optional heart-rate variability—and selects the next track to match that state on Russell’s valence–arousal circumplex. All signal processing runs in the browser; no audio, video, or physiological data leaves the device, and only anonymous self-report ratings are exported. We describe the affect model, the track-ranking rule, and a blinded, counterbalanced within-subject protocol that contrasts Vibe Shuffle against a Random Shuffle baseline drawn from the same fixed pool, with perceived mood-fit as the primary outcome and liking as a control. The system and its evaluation protocol are openly available, providing a reproducible, privacy-preserving testbed for state-aware music selection.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems*.

Keywords

affective computing, music recommendation, valence–arousal, facial expression, heart-rate variability, on-device processing

1 Introduction

People routinely use music to maintain or change how they feel, and the emotional fit of a track is a strong driver of listening satisfaction [2]. Streaming services recommend at enormous scale, but most recommenders optimise for long-run engagement inferred from listening history rather than for the listener’s *current* affective state. As a result, the same playlist is served whether a listener is tense and restless or calm and content.

A growing body of work in affective computing estimates affect from facial expression and from autonomic signals such as heart-rate variability, and organises it along the two dimensions of valence and arousal—a low-dimensional *core affect* rather than discrete, labelled emotions [5, 6]. Bringing such estimates to bear on music selection is attractive, but two obstacles recur: continuously streaming camera and physiological data to a server raises serious privacy concerns, and claims of “mood-aware” selection are rarely evaluated against a matched baseline.

We present *Vibe Shuffle*, a client-only web application that closes this loop entirely on-device. Facial expression is mapped to a continuous valence, an optional Bluetooth heart-rate sensor contributes arousal, the two are fused onto the valence–arousal circumplex, and the next track is chosen to match the fused state from a fixed, curated pool. No audio, video, or physiological data leaves the browser; only anonymous self-report ratings are exported.

Our contribution is twofold: a privacy-preserving, fully in-browser affect-to-music pipeline, and a blinded, counterbalanced validation protocol that contrasts Vibe Shuffle against a Random Shuffle baseline drawn from the same pool, isolating perceived mood-fit from mere liking. The system, the curated pool, and the export schema are openly available, making the design reproducible and falsifiable.

2 Methodology

2.1 System overview

Vibe Shuffle is a React/Vite single-page application with no backend. The signal chain runs client-side: the camera feeds a facial valence estimator, an optional heart-rate sensor feeds an arousal estimator, the channels are fused into a (*valence, arousal*) state, that state is mapped to one of four quadrants, and a ranking step selects the next track. Playback is handled by the Spotify Web Playback SDK so that licensed audio plays in the browser without the application ever receiving the audio stream. Throughout, the pipeline estimates a continuous valence–arousal coordinate—a behavioural-physiological proxy for *core affect* [6]—rather than classifying discrete, named emotions.

2.2 Valence from facial expression

A MediaPipe Face Landmarker [3] produces per-frame blendshape activations that are reduced to three facial-*expression* scores we label *happy*, *tense*, and *sad*—measures of facial display, not of inferred feeling. A slowly learned personal neutral baseline is subtracted to correct for resting facial morphology, and the positive-versus-negative balance is mapped to a continuous valence $v \in [0, 1]$. Head and body motion (nose-tip drift and frame-to-frame differencing) feed a motion channel that nudges the positive score upward and contributes to arousal.

2.3 Arousal from heart rate

When a Bluetooth heart-rate sensor is connected, heart rate (HR) and inter-beat (RR) intervals are parsed and a 120 s personal baseline is built. Short-window arousal combines the z -scored heart rate (higher HR raising arousal) with the z -scored RMSSD, a standard short-term heart-rate-variability index whose increase indicates



This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

Table 1: Track categorisation. Each track’s quadrant is fixed from its Spotify valence v_t and energy a_t by splitting both axes at 0.5; the pool holds 25 tracks per quadrant.

Quadrant	Rule
Energetic	$v_t \geq 0.5, a_t \geq 0.5$
Calm	$v_t \geq 0.5, a_t < 0.5$
Tense	$v_t < 0.5, a_t \geq 0.5$
Melancholic	$v_t < 0.5, a_t < 0.5$

lower arousal [7]:

$$a_{\text{phys}} = \text{clip}(\frac{1}{2} + \alpha z(\text{HR}) - \beta z(\text{RMSSD}), 0, 1). \quad (1)$$

SDNN is logged for signal-quality control but excluded from the short-window estimate. The export also retains a rhythm diagnostic for exploratory analysis. It is not used as the arousal driver because it measures regularity rather than baseline-relative autonomic activation. The sensor is strictly optional; the system degrades gracefully without it.

2.4 Fusion and quadrants

The fused state takes valence from the face and arousal primarily from the heart, with head motion m added on top:

$$(v, a) = (v_{\text{face}}, a_{\text{phys}} + \gamma m). \quad (2)$$

A usable electrocardiogram sets the arousal base in both directions; without one, the face/motion channel carries arousal upward only; with neither face nor sensor, both axes default to the neutral midpoint 0.5. Splitting each axis at 0.5 yields four quadrants—Energetic, Calm, Tense, and Melancholic—consistent with the circumplex model of affect [5, 6].

2.5 Music pool

The pool is a fixed set of 100 tracks built once from the public Kaggle *Spotify Tracks Dataset* of real Spotify audio features [4]. Every track carries Spotify’s two affect-bearing features on $[0, 1]$: *valence*, the musical positivity of a track, and *energy*, its perceived intensity and activity. We take valence as the track’s valence coordinate and energy as its arousal coordinate, so each track t inherits a fixed (v_t, a_t) with no per-session lookup. Tracks are categorised by the same midpoint split used for the listener (Table 1): *Energetic* when $v_t \geq 0.5$ and $a_t \geq 0.5$, *Calm* when valence is high but energy low, *Tense* when valence is low but energy high, and *Melancholic* when both are low. To assemble a balanced pool, the most popular, de-duplicated tracks with Spotify popularity ≥ 55 were taken from each quadrant until 25 were collected, yielding 100 tracks split evenly across the four quadrants. The same pool is used for every participant, which keeps the Vibe-versus-Random comparison controlled and removes any dependence on a participant’s personal library.

2.6 Track selection

Each track’s quadrant is fixed in advance from its (v_t, a_t) , so selection reduces to a draw over a candidate set. The fused listener state (v^*, a^*) is mapped to its quadrant q^* . In a Vibe block the next track

Table 2: Selected exported fields (one row per played track).

Field	Meaning
block_mode	condition: vibe or random
song_quadrant	target affect quadrant of the track
detected_valence	fused listener valence
detected_arousal	fused listener arousal
physiology_arousal	baseline-relative HR/RMSSD arousal estimate
physiology_coherence	exported cardiac-coherence diagnostic, not a selection driver
physiology_quality	heart-rate signal-quality flag
rating_fit_1_to_7	primary outcome: mood-fit
rating_like_1_to_7	control: liking

is drawn from the mood-congruent quadrant only,

$$t^* \in \{t : \text{quad}(t) = q^*\}, \quad (3)$$

with recently played tracks deprioritised so a block does not repeat songs. Within that set the specific track is fixed by a per-session seed, making each run reproducible yet different. A Random block applies the same seeded rule to the entire pool. The two conditions thus share an identical pool and an identical random mechanism; the *only* manipulated variable is the quadrant constraint. For transparency the Euclidean distance between the detected state and each track is also computed and shown to the listener as a “fit” percentage, but it does not determine which track plays.

2.7 Validation protocol

The application implements a blinded, counterbalanced within-subject comparison. A session contains two blocks—random and vibe—of five tracks each drawn from the shared pool of Table 1; block order is randomised per session and recorded for counterbalancing. Each track plays for 60 s before the rating prompt opens, and the condition is never revealed during the session. After each track, two 5-point questions are asked in sequence: *liking* (“How much do you like this song?”) and *mood-fit* (“How well did it fit your current mood?”). Mood-fit is the primary outcome; liking is the control, separating “did not fit my mood” from “I simply do not like this song.”

3 Results

This paper reports the implemented system and its pre-registered evaluation design; a participant study using the protocol above is the immediate next step, and no participant data is reported here. The working artifact demonstrates that the full affect-to-music loop runs in real time, in-browser, on commodity hardware with only a webcam and an optional consumer heart-rate strap.

Each completed session exports one row per track capturing the song features, the detected (v, a) state, signal-quality flags, and both ratings (Table 2). The pre-specified analysis contrasts the mood-fit rating between vibe and random blocks, controlling for liking and accounting for block order; liking is expected to be comparable across conditions (both blocks draw from the same pool), while mood-fit is the discriminating measure.

4 Discussion

By keeping the entire pipeline on-device, Vibe Shuffle sidesteps the privacy cost that usually accompanies camera- and physiology-driven personalisation: raw frames and heart-rate packets are processed in memory and discarded, and only the anonymous rating CSV ever leaves the browser, at the participant’s initiative. This makes the system suitable for low-friction field deployment and for classroom or seminar replication.

Several limitations bound the claims. The system does not recognise discrete emotions: it estimates a low-dimensional core-affect proxy from a facial-expression display, autonomic arousal, and head motion. Facial movements are not a reliable read-out of felt emotion—the same expression recurs across states and the same state across expressions [1]—and physiological arousal is undifferentiated and valence-blind, raised alike by excitement, exertion, or the played music itself; the detected state is therefore a behavioural-physiological correlate of affect rather than affect itself. Because both axes are referenced to a per-session personal baseline, the system tracks change relative to the listener’s starting point rather than an absolute state: a listener who begins aroused has that state absorbed into the neutral baseline, so the absolute initial affect is unobserved—a self-reported baseline (e.g. a Self-Assessment Manikin) would anchor it. The estimator is a transparent heuristic rather than a validated clinical affect measure; facial valence is sensitive to lighting and occlusion, and the arousal channel depends on a compatible sensor that not every participant will own. The pool is small and fixed by design, so results speak to relative ranking *within* that pool rather than to catalogue-scale recommendation. Finally, self-reported mood-fit is a perceptual proxy; it captures whether selection *feels* apt, which is the intended user experience, but not an objective change in affective state.

5 Conclusion

Vibe Shuffle shows that a privacy-preserving, fully in-browser system can estimate a listener’s valence and arousal and select music to match it, and it pairs that system with a blinded, counterbalanced protocol that can falsify the resulting “mood-aware” claim against a matched random baseline. The artifact and protocol are openly available, providing a reproducible testbed; running the pre-registered study to estimate the mood-fit advantage of state-aware selection is the natural next step.

References

- [1] Lisa Feldman Barrett, Ralph Adolphs, Stacy Marsella, Aleix M. Martinez, and Seth D. Pollak. 2019. Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements. *Psychological Science in the Public Interest* 20, 1 (2019), 1–68. doi:10.1177/1529100619832930
- [2] Patrik N. Juslin and Daniel Västfjäll. 2008. Emotional Responses to Music: The Need to Consider Underlying Mechanisms. *Behavioral and Brain Sciences* 31, 5 (2008), 559–575. doi:10.1017/S0140525X08005293
- [3] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. *arXiv preprint arXiv:1906.08172* (2019).
- [4] maharshipandya. 2022. Spotify Tracks Dataset. Kaggle. <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>.
- [5] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The Circumplex Model of Affect: An Integrative Approach to Affective Neuroscience, Cognitive Development, and Psychopathology. *Development and Psychopathology* 17, 3 (2005), 715–734. doi:10.1017/S0954579405050340
- [6] James A. Russell. 1980. A Circumplex Model of Affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178. doi:10.1037/h0077714
- [7] Fred Shaffer and J. P. Ginsberg. 2017. An Overview of Heart Rate Variability Metrics and Norms. *Frontiers in Public Health* 5 (2017), 258. doi:10.3389/fpubh.2017.00258

Character Count

This work contains 10128 characters (without spaces).